

Current-host helicity-summed comparison (n=2..9)

IN PROGRESS SNAPSHOT | absent cells are unattempted

--fft | recurrence: persisted all-helicity physical schedule | OTF compact artifact: all runtime helicities

pyAmpliCol setup: generation + fresh load + first complete helicity sum; OTF includes family warm-up

Reference setup: build/init/first pass | AmpliCol setup: process/raw-library generation/build + immutable snapshot

Warmed runtime measured separately | compiled FFT disabled

pyAmpliCol CLI profile: batch 128 (cyclic 10 points) | Reference/AmpliCol: scalar aggregates normalized per point

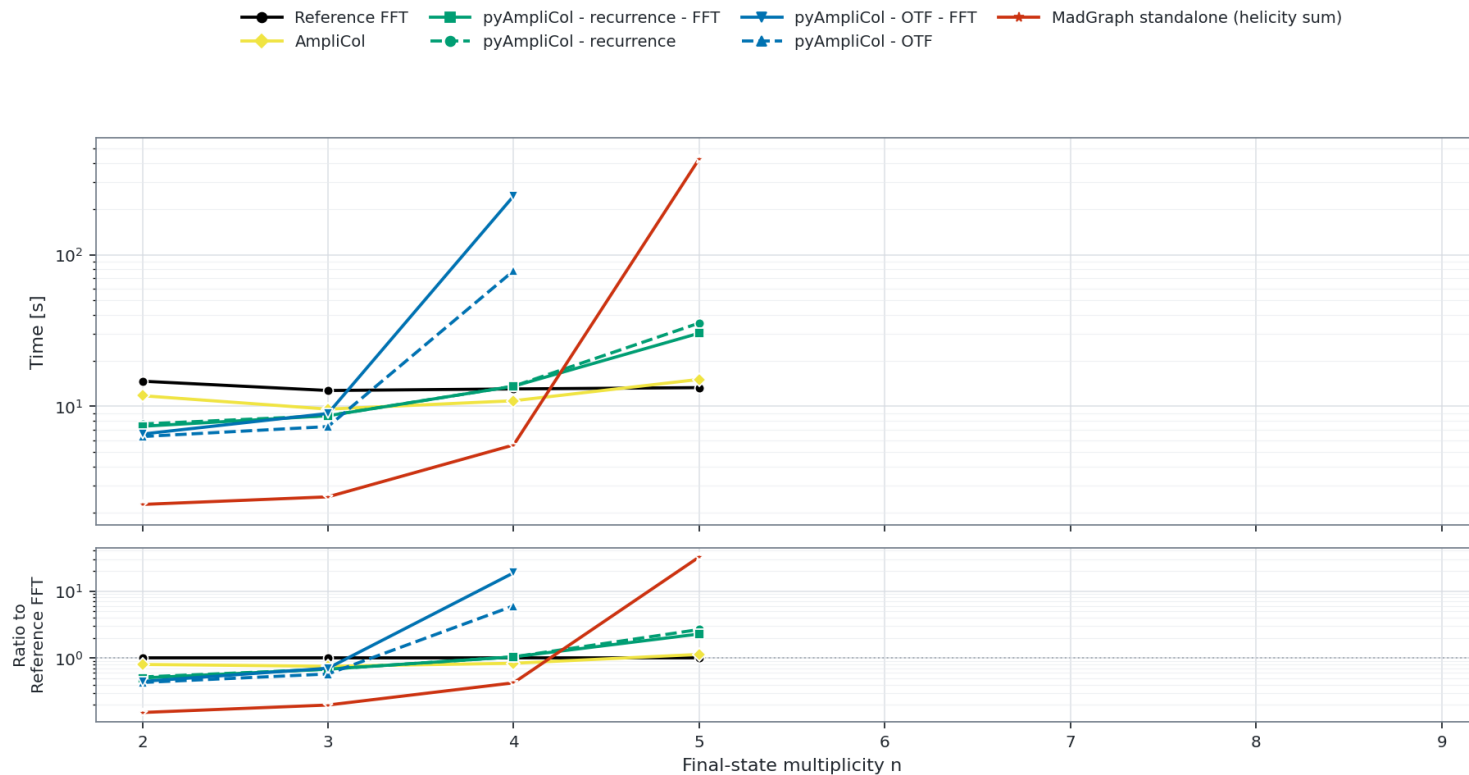
AmpliCol: create-raw bulk H family + probe-local zero pruning | Reference FFT: analytic-nonzero H sweep

FullColor Setup time: $gg \rightarrow gg + (n - 2)g$

Campaign --fft enabled | artifacts: all runtime helicities | setup/warm workload: complete helicity sum

AmpliCol: create-raw bulk H family | pyAmpliCol CLI profile: batch 128 (cyclic 10 points)

Reference/AmpliCol: scalar aggregates normalized per point | compiled FFT disabled | log-scale Y axes



Failures and omitted points

- pyAmpliCol - OTF: n=5 failed (5 min setup-time limit).
- pyAmpliCol - OTF - FFT: n=5 failed (5 min setup-time limit).
- Campaign is still running; absent cells are unattempted and omitted.
- Sparse ddbar n=6 helicity-sum extension uses a 30-GiB per-cell guard; its immutable source policy is recorded per overlaid cell.
- Measured ddbar n=6 OTF extensions use a 10-hour setup/runtime cap and a 30-GiB process-tree guard; exact immutable source policies are recorded per overlaid cell.
- MadGraph standalone uses generated SMATRIX with USERHEL=-1; warmed GOODHEL pruning remains enabled.
- MadGraph points are retained from a same-workstation snapshot that predates the node-fingerprint field; strict terminal host authentication is therefore unavailable for this anytime render.

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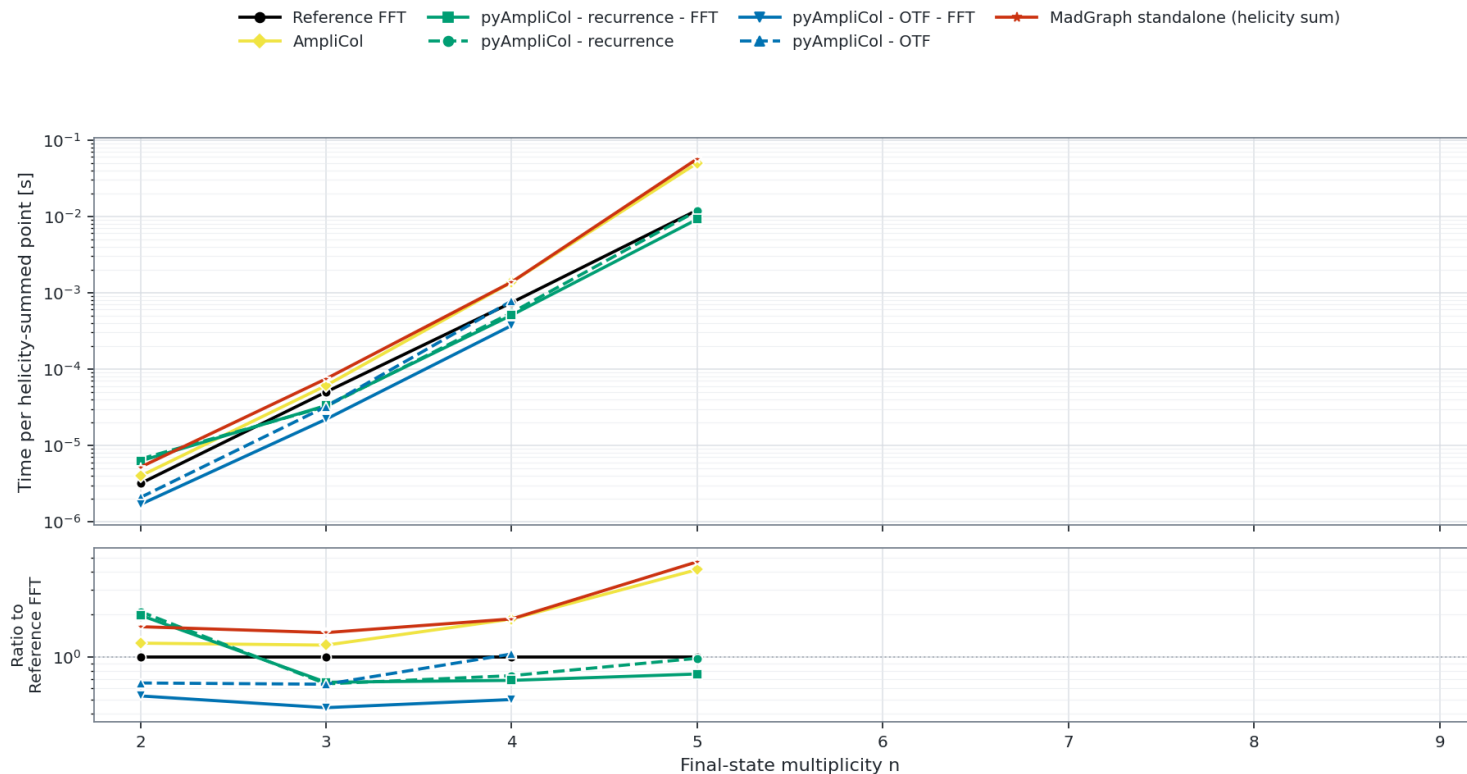
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FullColor Warmed helicity-summed runtime per point: $gg \rightarrow gg + (n-2)g$

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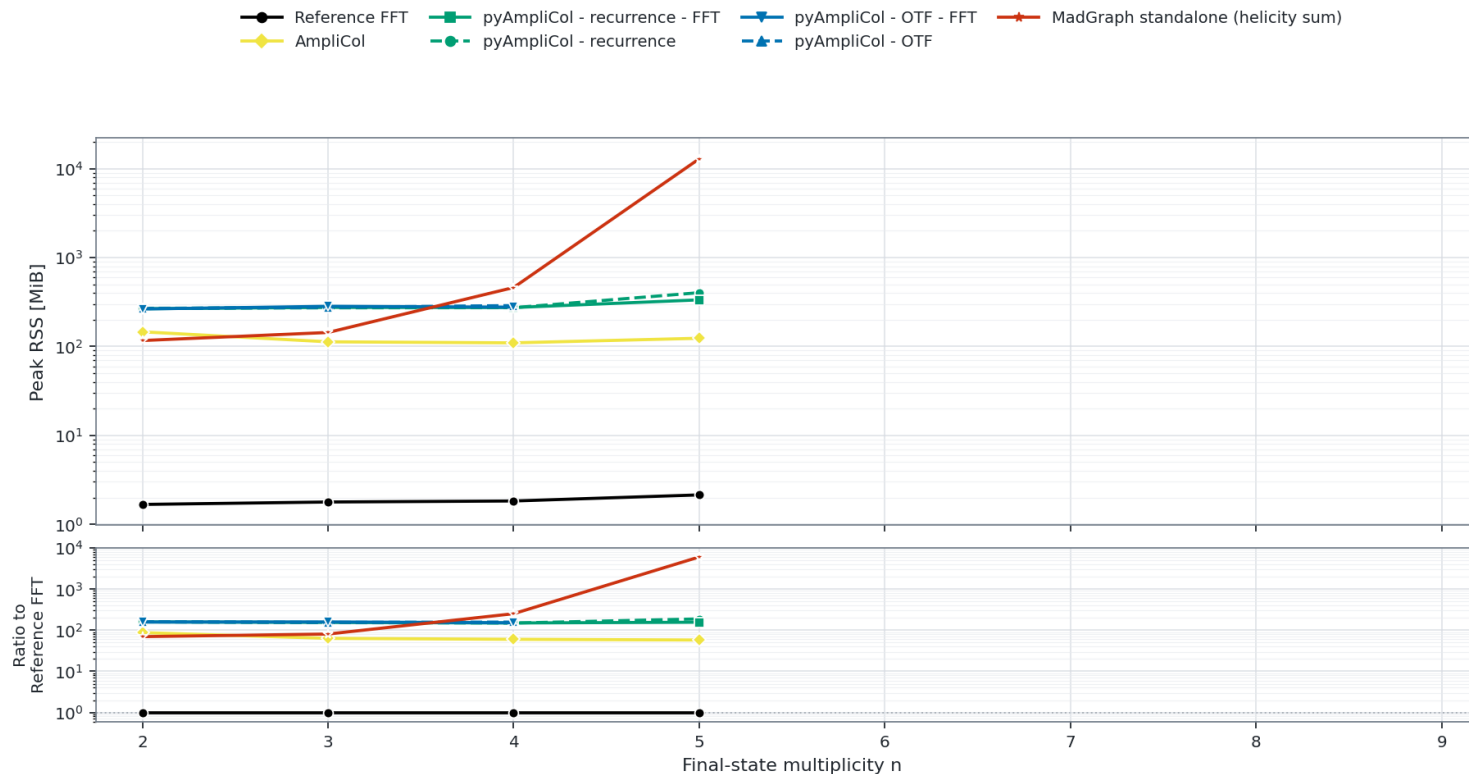
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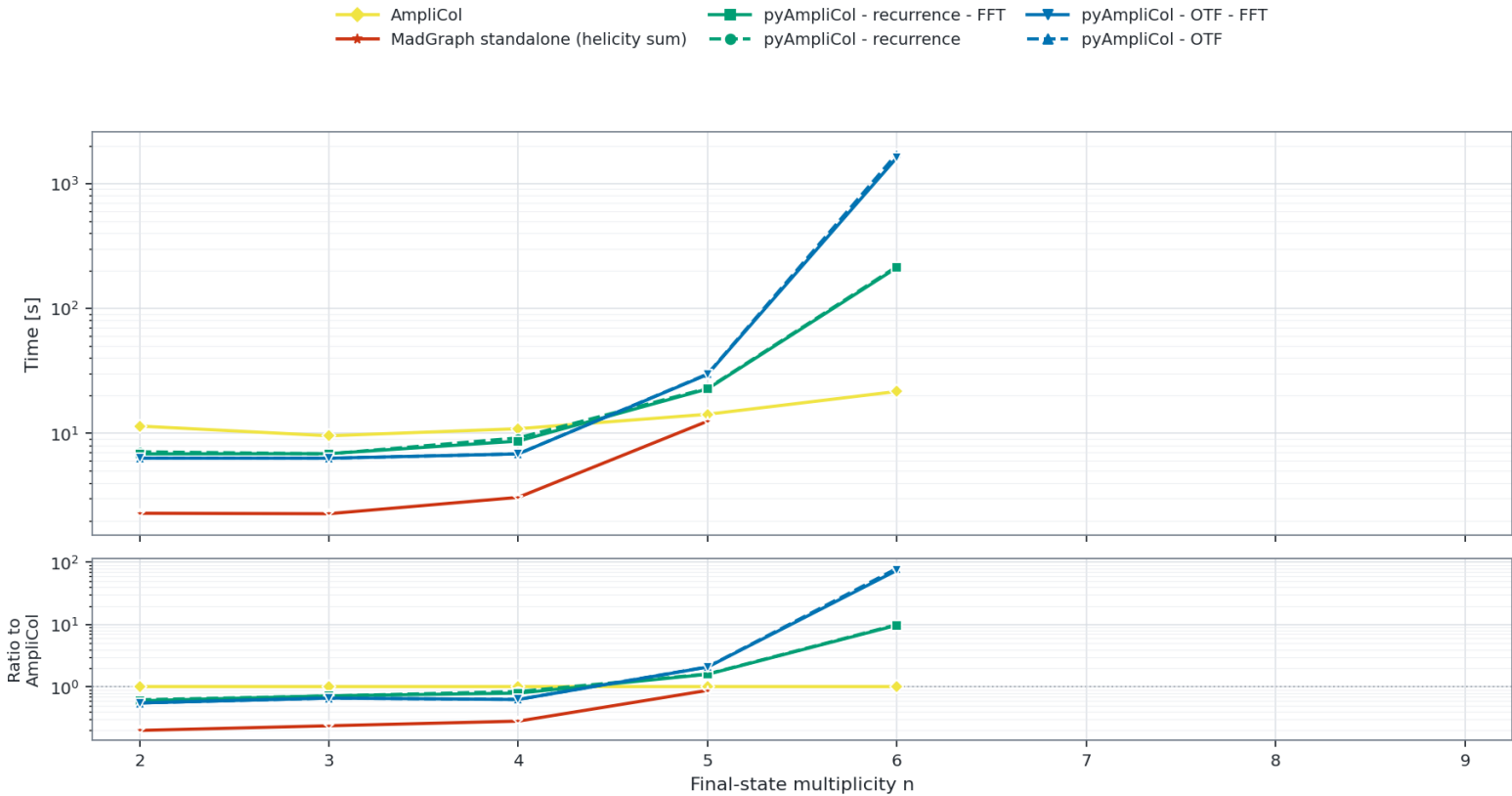
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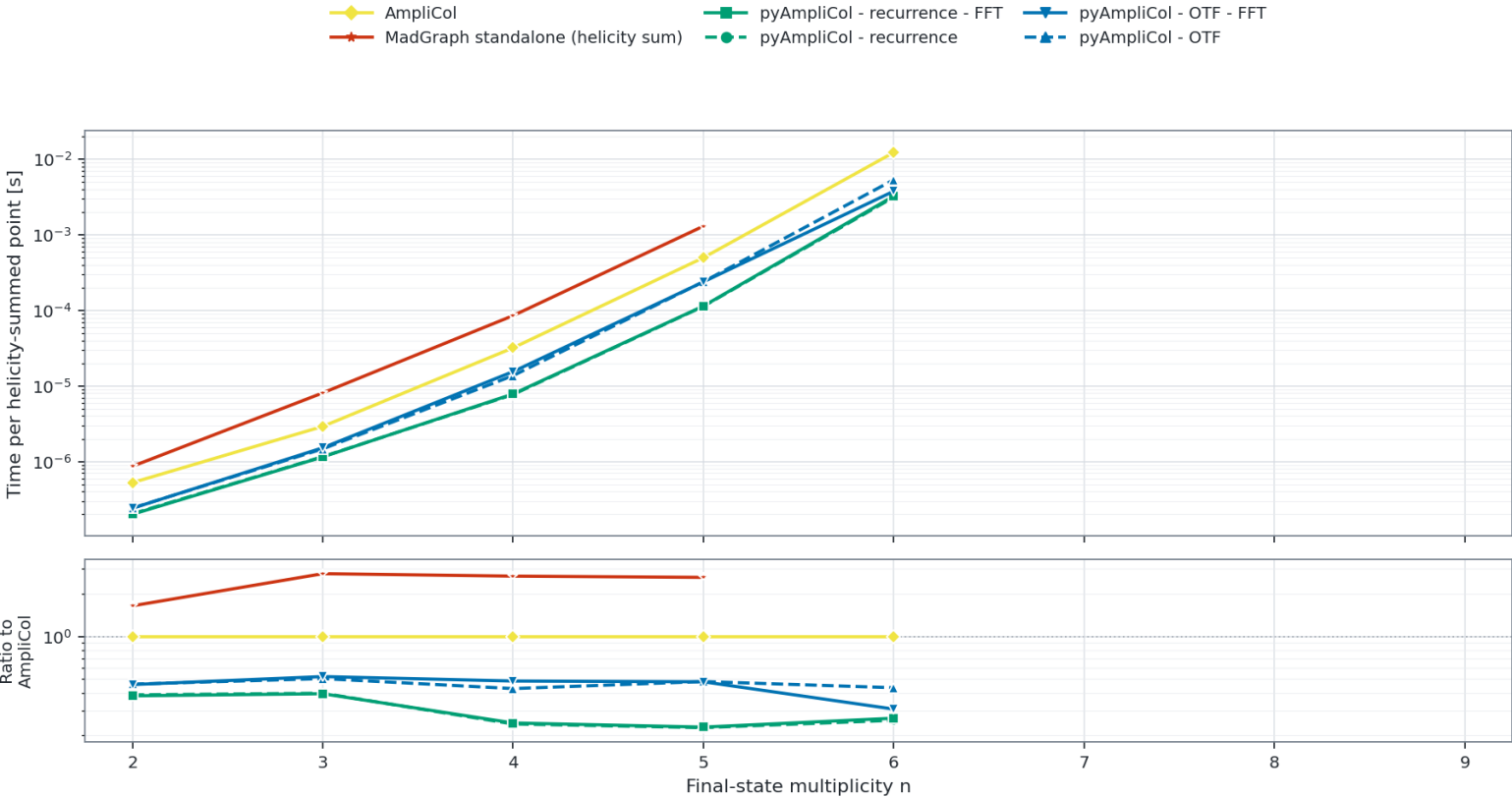
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FullColor Warmed helicity-summed runtime per point: $d\bar{d} \rightarrow d\bar{d} + (n - 2)g$

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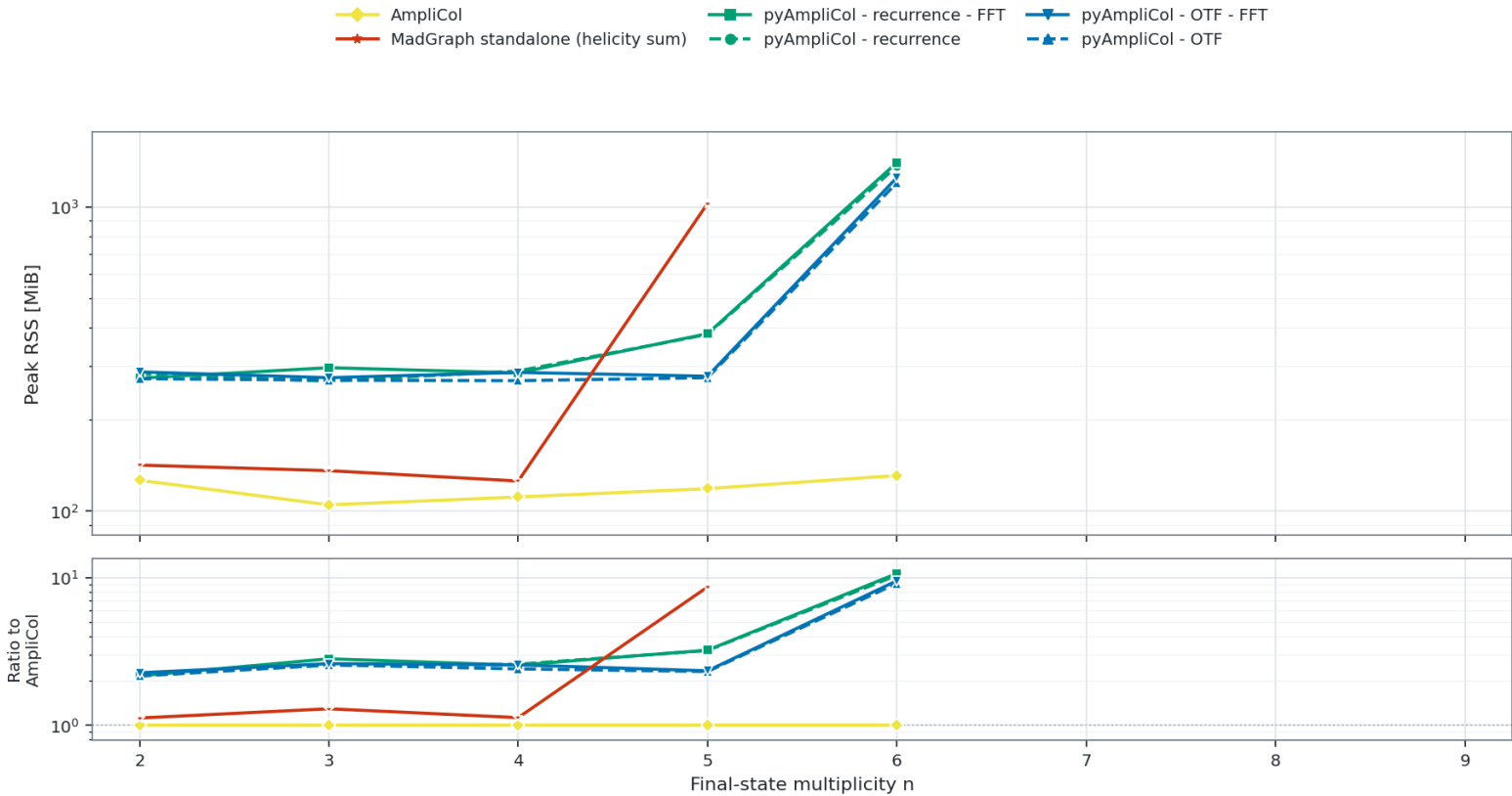
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